



Online Fixed Acoustic Camera

User Manual & API

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1 System Introduction

1.1 Overview

An acoustic camera, also known as an acoustic imager, is a device capable of converting sound fields into visual images. Acoustic imaging utilizes microphone array measurement technology. By measuring the phase differences of sound wave signals arriving at each microphone within a specific space, it determines the location of sound sources and measures their amplitude based on phased array principles. This process displays the spatial distribution of sound sources as an image—forming a spatial acoustic field distribution map, known as a sonogram. In the sonogram, the image's color and brightness represent the intensity of sound. This device features visual capabilities for sound acquisition, image capture, and sound localization, enabling visual inspection and diagnosis of equipment failures. Compact and lightweight, it can be mounted on various inspection robots, significantly enhancing their inspection capabilities.

This system employs internationally leading ESR (Environmental Sound Recognition) technology for target identification and beamforming-based sound source localization technology to measure the position of sounds emitted by equipment and the state of sound radiation. This technology visualizes sound across the entire spectrum from audible to ultrasonic frequencies through color-coded contour maps. It effectively measures acoustic field distributions, seamlessly overlaying acoustic maps onto visible light video imagery—similar to how thermal imaging cameras detect object temperatures. Beyond traditional acoustic camera functions—including sound capture, image acquisition, and acoustic localization visualization—this device enables real-time detection and pinpointing of

anomalies during equipment malfunctions.

It serves diverse industrial applications, including but not limited to: partial discharge monitoring in high-voltage transmission lines, gas leak detection in high-pressure pipelines, and partial discharge fault monitoring in transformers and switchgear. Additionally, the system supports sound source localization, anomaly sound detection, and sound source tracking. It delivers exceptional detection performance for both steady-state and highly transient sound sources, as well as stationary or moving objects.



Figure 1-1 Partial Discharge Monitoring Scenario



Figure 1-2 Gas Leak Monitoring Scenario



Figure 1-3 Anomaly Sound Monitoring Scenario

1.2 System Features

Compared to other types of partial discharge monitoring, gas leak detection, or abnormal sound monitoring products, the airborne acoustic imager offers the following characteristics:

- Integrated intelligent acoustic-optical imaging design with a compact form factor;
- Intrinsically safe explosion-proof design compliant with explosion-proof requirements;
- Compatible with infrared modules for integrated infrared detection capabilities.
- Wide detection frequency range covering both audible and ultrasonic frequencies
- Automated acoustic field visualization for intuitive, rapid, and precise localization
- Supports steady-state continuous detection and transient detection modes
- Capable of imaging common partial discharge signal types
- Capable of imaging gas leaks in high-voltage pipelines
- Open interfaces and protocols supporting secondary development

2 Product Specifications

2.1 External Dimensions

Figure 2-1 shows the actual physical appearance of the product.



Figure 2-1 Product Appearance

Figure 2-2 shows the product's external dimensions.

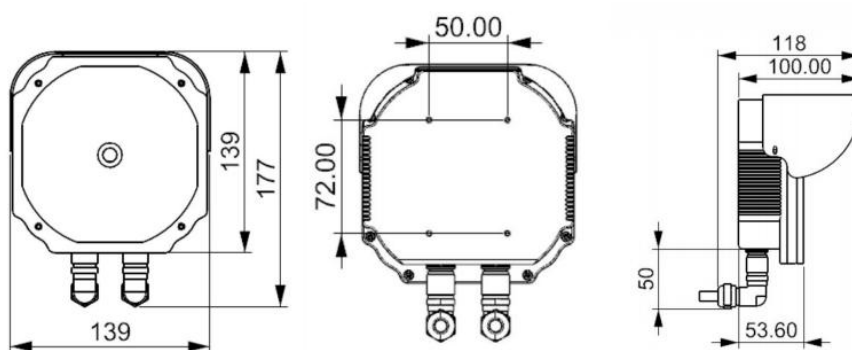
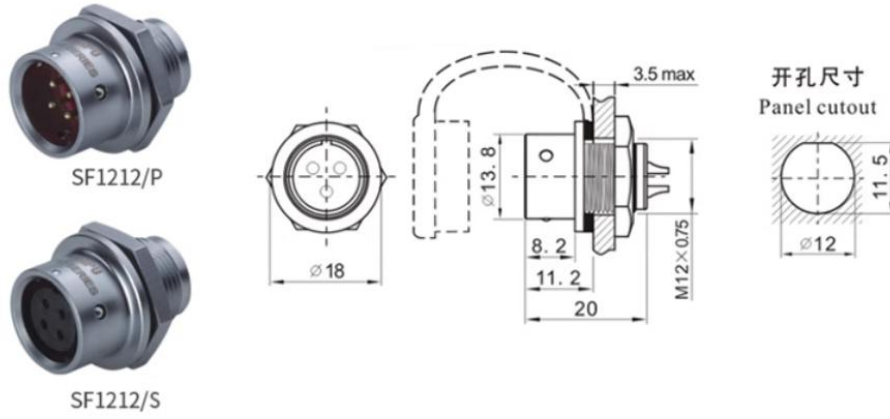


Figure 2-2 Product External Dimensions

SF1212/P SF1212/S



SF1214/P SF1214/S

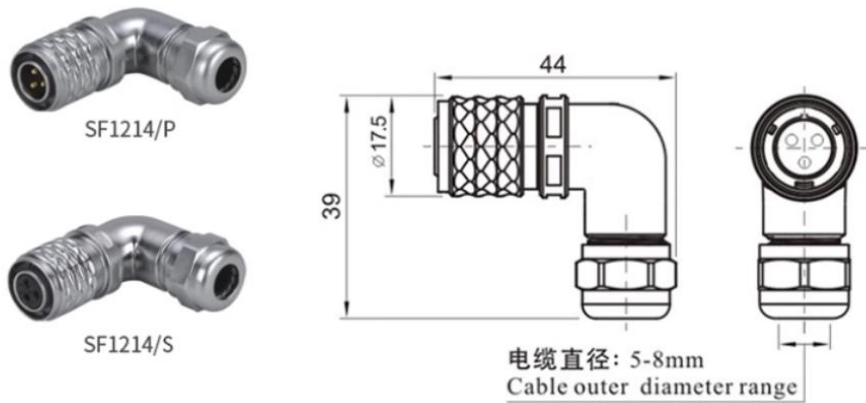


Figure 2-3 Row Spacing Dimensions

2.2 Performance Specifications

Parameter Type	Parameter Specification
Model	HZ-FA-371
Acoustic Imaging Parameter	
Acoustic Sensors Number	128 Unit
Acoustics Sensors Type	MEMS Digital Microphone
Sensor Sensitivity	-26±1 dBFS (1kHz Sensitivity @1Pa)
SPL Range	-30dB~120dB
Frequency Range	2kHz~65kHz
Min Sensitivity (1m)	5dB SPL 18KHz 7dB SPL 30KHz 11dB SPL 42KHz 14dB SPL 60KHz
Multiple-Target	Support
Imaging Frame	≥30 FPS
Detection Distance	0.3m~130m
Typical Application	Partial Discharge Detection, Gases Leakage Detection, Mechanical Noise Detection
Visible Light Camera	
Camera Pixel	13MP
FOV	72°±3°
Video Resolution	1920*1080@30fps

Electrical Parameter	
Rated Input	DC12V 3A
Operating Power Consumption	$\leq 9W@25^{\circ}C$
Weight	<0.8 kg
Data Transmission	Ethernet
Environmental Adaptability	
Operating Temperature	-20°C~+55°C
Storage Temperature	-30°C~+70°C
Humidity	10% ~ 95%, Non-Condensing
Protection Level	Greater than IP65
Explosion-Proof	ExibI ICT6Gb; ExibII ICT80°C Db
EMC Reliability Indicators	Electrostatic Discharge Immunity, GB/T17626.2-2018 Level 4, Class A Power Frequency Magnetic Field Immunity, GB/T17626.8-2006 Level 4, Class A Pulsed Magnetic Field Immunity, GB/T17626.9-2011 Level 4, Class A
Others	
Multiple-Language	Simplified Chinese, English
Audio Calibration	Support

3 System Configuration

3.1 Equipment Installation

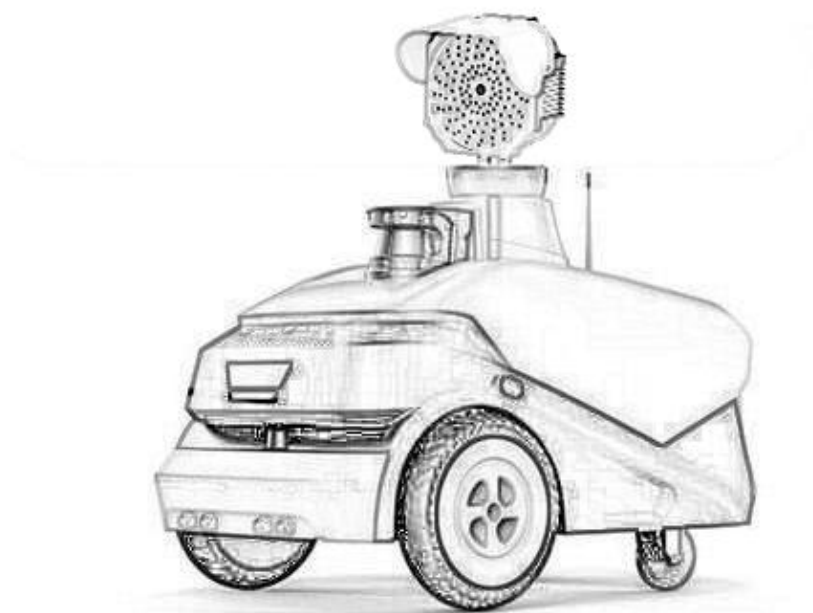


Figure 3-1 Equipment Installation

This compact and lightweight device can be mounted on a robotic gimbal or other suitable locations (as shown in the figure above). During installation, ensure no objects obstruct the front of the device and minimize the impact of robotic vibrations to prevent compromising subsequent detection performance. The device's orientation may be adjusted according to robotic inspection requirements. Structural components required for installation must be provided by the user.

3.2 Network Topology

Figure 3-2 shows the network connection topology for devices in a typical scenario. Devices and computers are generally located within the same local area network (LAN). Through the HTTP protocol, real-time video feeds from the field can be transmitted back, and parameter configurations can be

performed.

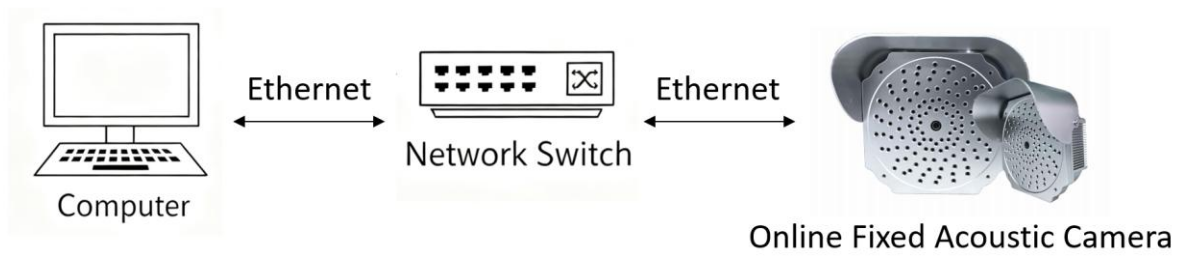


Figure 3-2 Network Connection Topology Diagram

Note: Default device IP: 192.168.2.204. The computer IP must be within the same subnet as the device IP.

4 System Configuration Overview

4.1 System Logging

With a normal network connection, enter the device IP address in your browser (default information is on the label on the back of the acoustic module) to access the login page, as shown in Figure 4-1. The default username and password are as follows:

Default username: admin Default password: 123456

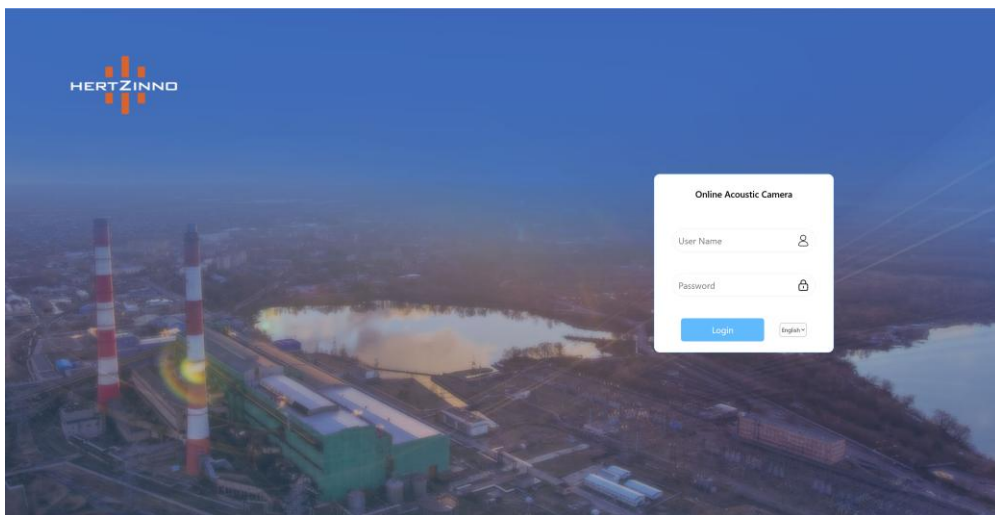


Figure 4-1 Logging Page

4.2 Preview

After logging in, you will enter the preview interface. The left side of the page displays the live video feed, while the right side houses the parameter control panel. The “Leak Detection” feature in the upper-right corner indicates the current device operating mode. The power button allows you to log out of the system.

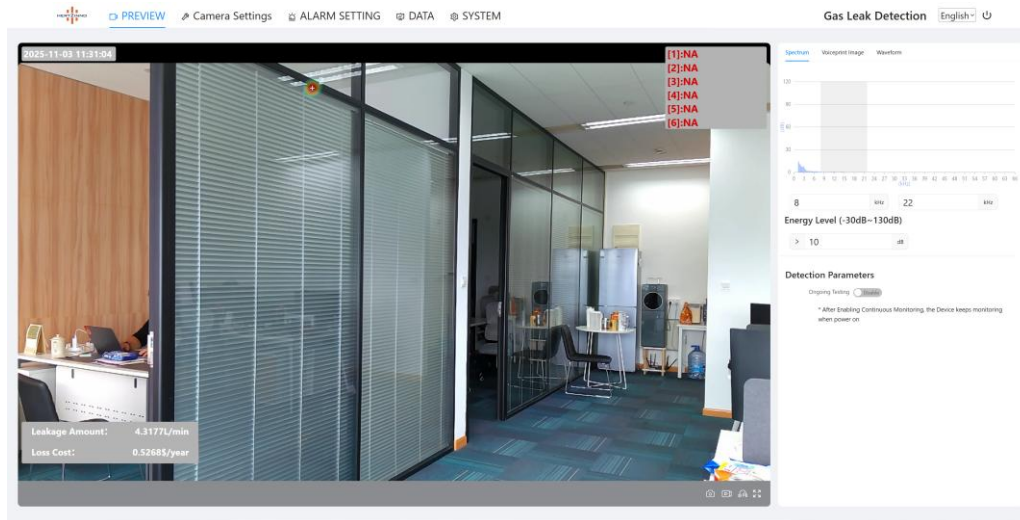


Figure 4-2 Real-time Interface

Real Time Video

This interface allows real-time viewing of the current video feed and adjustment of parameters on the right-hand control panel. The system time is displayed in the upper-left corner of the video, while the upper-right corner shows the real-time environmental sound energy within the monitored frequency band. When the environmental energy level within the corresponding frequency band exceeds the threshold value, an anomaly will be indicated, and a sound field diagram will appear on the live video feed. The location indicated by the sound field diagram corresponds to the air leak.

Control Pannel

The functions of each section of the control pannel are described as follows:

- **Spectrum Plot:** Performs real-time analysis and plots the frequency spectrum of the current ambient sound signal. The spectrum plot provides a visual representation of the distribution of sound across different frequency bands in the environment, serving as a reference for adjusting imaging frequency bands.

- Acoustic Fingerprint: Real-time analysis and visualization of the time and frequency characteristics of the current environmental sound signal.
- Waveform: Real-time analysis and visualization of the temporal variation of the current environmental sound signal, intuitively reflecting changes in signal amplitude, phase, and other parameters over time.
- Frequency Range: Used to adjust the current monitored sound frequency range. The adjustable range is 2~65 kHz, and the input value must be an integer. The difference between upper and lower limits must not exceed 20 or fall below 2, and the upper limit must exceed the lower limit. After band adjustment, the shaded area in the spectrum diagram indicates the currently selected frequency band range.
- Energy Level: Adjusts the detection threshold value. When the ambient energy level exceeds the threshold, the status changes to abnormal, enabling real-time acoustic field imaging. The video displays an acoustic field imaging marker.

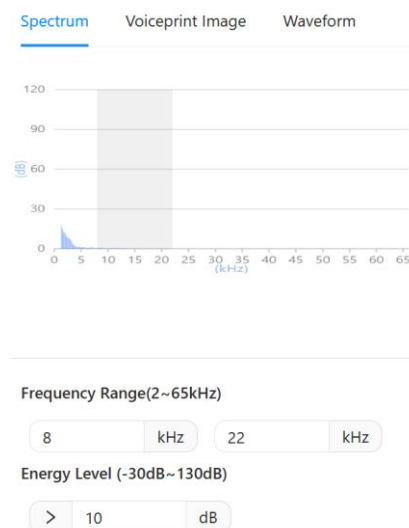


Figure 4-2-1 Console Interface

- **Single Detection:** Used for on-site inspection and recording results. Once inspection begins, a normal point information is logged in real time and corresponding data is generated for viewing on the data review page. (Enables single-run inspection mode)

Energy Level (-30dB~130dB)

> 10 dB

Start Detection

Latest Result

No Test Result Yet

Figure4-2-2 Single Detection

- **Intelligent Cruise:** Used for cyclic inspections, triggering alerts upon detecting anomalies and recording inspection results. After continuous patrols, information on abnormal points is recorded in real time, generating corresponding data viewable on the data viewing page. (Enable Smart Patrol Mode).

Energy Level (-30dB~130dB)

> 10 dB

Detection Parameters

Ongoing Testing ☐ Disable

* After Enabling Continuous Monitoring, the Device keeps monitoring when power on

Figure4-2-3 Intelligent Cruise

4.3 Camera Settings

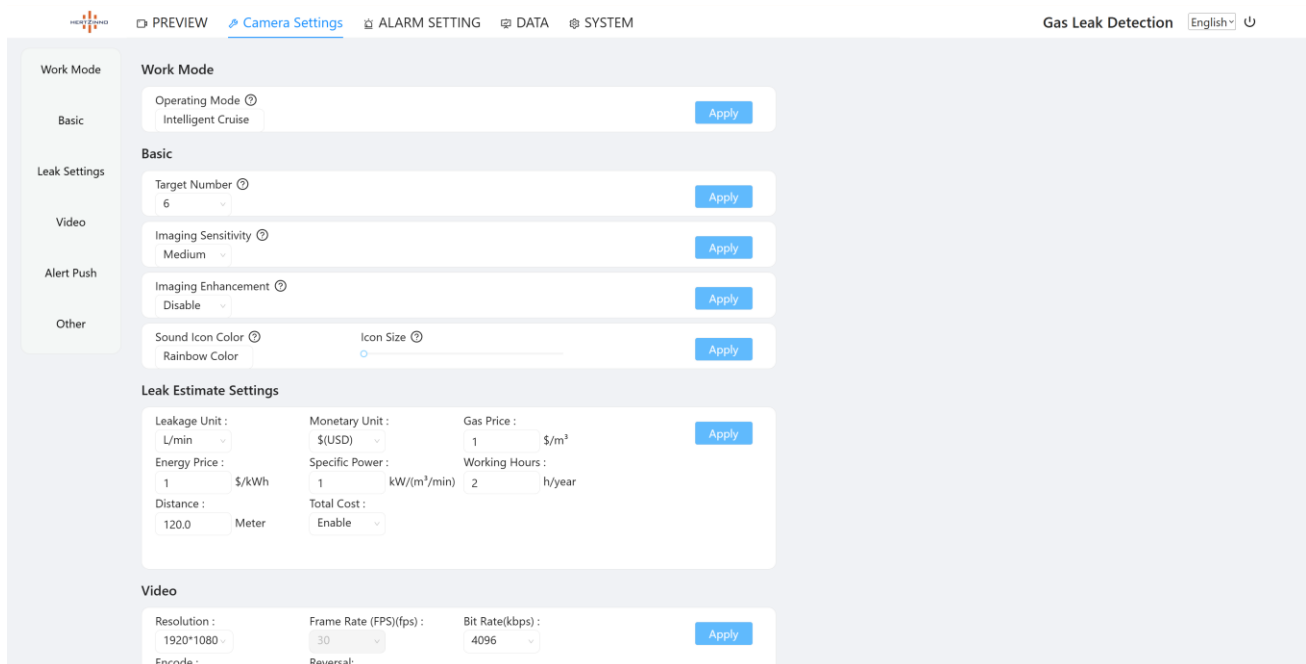


Figure 4-3 Camera Settings

Work Mode

The device operation mode can be set to single detection or intelligent cruise .

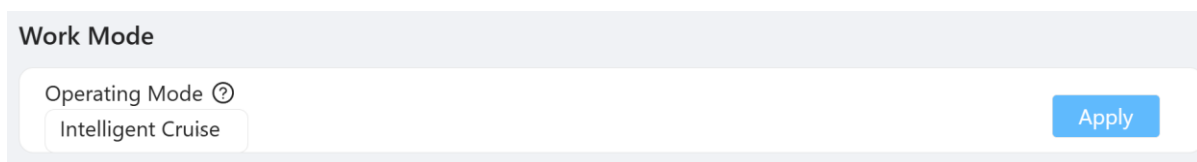
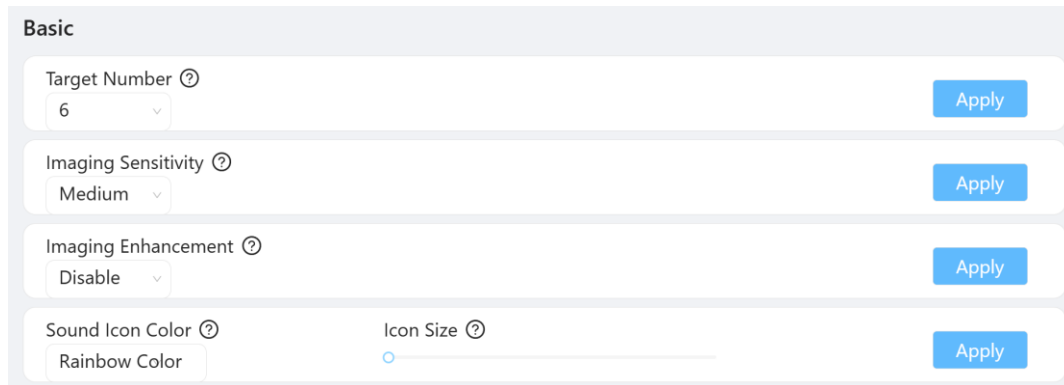


Figure 4-3-1 Work Mode Settings

- **Single Detection Mode:** This mode initiates detection based on on-site requirements. After completing a single detection cycle, it pauses until the next trigger occurs.
- **Intelligent Cruise Mode:** When enabled, this mode automatically cycles through configured alarm detection strategies without requiring manual retriggering.

Basic



The screenshot shows a 'Basic' settings panel with the following controls:

- Target Number**: A dropdown menu currently showing '6'.
- Imaging Sensitivity**: A dropdown menu currently showing 'Medium'.
- Imaging Enhancement**: A dropdown menu currently showing 'Disable'.
- Sound Icon Color**: A dropdown menu currently showing 'Rainbow Color'.
- Icon Size**: A horizontal slider control.

Each of these four rows has a blue 'Apply' button to its right.

Figure4-3-2 Basic

- **Target Number**: Refers to the maximum number of target points within the acoustic field that the system can simultaneously mark and output location information for in real time.
- **Imaging Sensitivity**: Indicates the system's responsiveness to acoustic signals. Higher sensitivity enhances real-time performance.
- **Imaging Enhancement**: Improves the stability of imaging points.
- **Sound Icon Color**: Allows customization of the color and size of the imaging point cloud map.

Video Settings

Configurable video resolution, video bitrate, reversal, etc. Video encoding defaults to H.264, video frame rate defaults to 30fps (non-modifiable); video resolution supports 1920*1080, 1280*720; Video bitrates support 4096, 2048, 1024, and 512; video flipping functions support horizontal mirroring, vertical mirroring, and center mirroring.

Video

Resolution :	Frame Rate (FPS)(fps) :	Bit Rate(kbps) :	Apply
1920*1080 ▾	30 ▾	4096 ▾	
Encode :	Reversal:		
H.264	Horizontal ▾		

Figure 4-3-3 Video Settings

Alert Push

It allows alert push and sets the pushed IP address.

Alert Push

Alarm Enable: ☐
Pushed address:

Apply

Figure 4-3-4 Alert Push

Leakage Settings

Once total cost is enabled, the estimation will be displayed in the screen.

Leak Estimate Settings

Leakage Unit :	Monetary Unit :	Gas Price :	Apply
L/min ▾	\$(USD) ▾	1 \$/m ³	
Energy Price :	Specific Power :	Working Hours :	
1 \$/kWh	1 kW/(m ³ /min)	2 h/year	
Distance :	Total Cost :		
120.0 Meter	Enable ▾		

Figure 4-3-5 Leakage Estimate Settings

Other

Other settings include Center Area Mode and Multiple Areas Mode. When Center Area Mode is enabled, the system will only detect abnormal sound points within the focus box area. Similarly, when Multiple Areas Mode is enabled, the system will only detect abnormal sound points within the custom-selected

area.

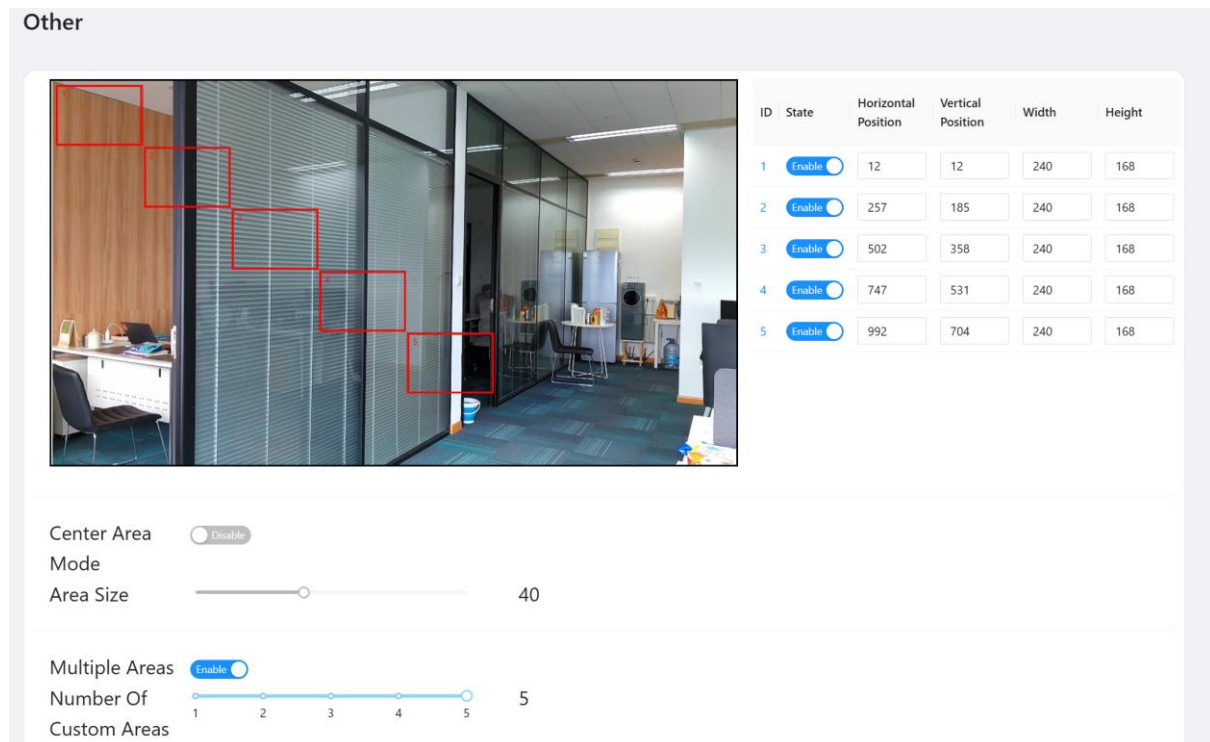


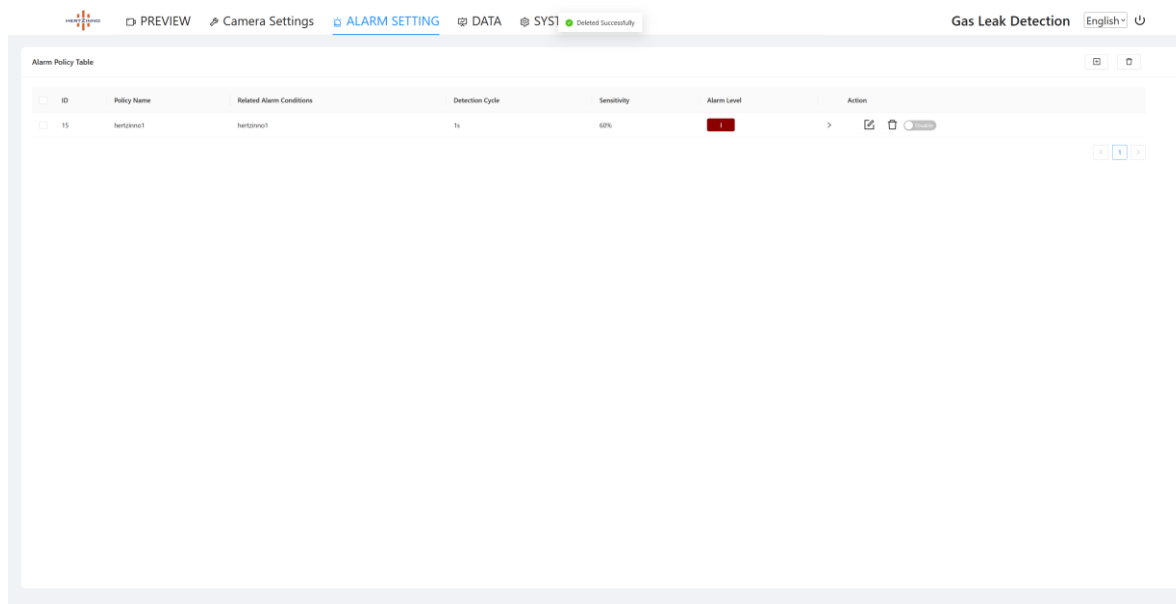
Figure 4-3-6 Other Settings

4.4 Alarm Settings

On this page, users can customize alert policies according to their needs. Custom alert policies include the following:

- Policy Name: The name of the policy
- Related Alarm Conditions: Includes decibel alerts, main frequency alerts, and spectrum comparison. Users can flexibly select and configure these alert conditions based on actual requirements to ensure accurate triggering under specific circumstances.
- Detection Cycle: The duration of a single detection, adjustable from 1 to 30 seconds. Users can modify the cycle length to balance detection accuracy with operational requirements.

- **Sensitivity:** This parameter measures the ratio of signals exceeding the detection threshold during a single scan. Higher sensitivity indicates greater responsiveness to anomalies but may increase false alarm rates. Users should set sensitivity appropriately based on actual conditions.
- **Alarm Level:** This field marks the urgency or importance of a policy. By setting different alarm levels, users can prioritize alerts of varying types, ensuring critical and urgent alerts receive priority handling when resources are limited.
- **Actions:** This field provides a means for users to document explanations regarding policy definitions. Users can input detailed descriptions here about the policy's background, purpose, applicable scenarios, and other relevant information for future reference and understanding. By filling in diagnostic notes, users can better convey the intent behind policy creation and facilitate other users' comprehension and application of the policy.



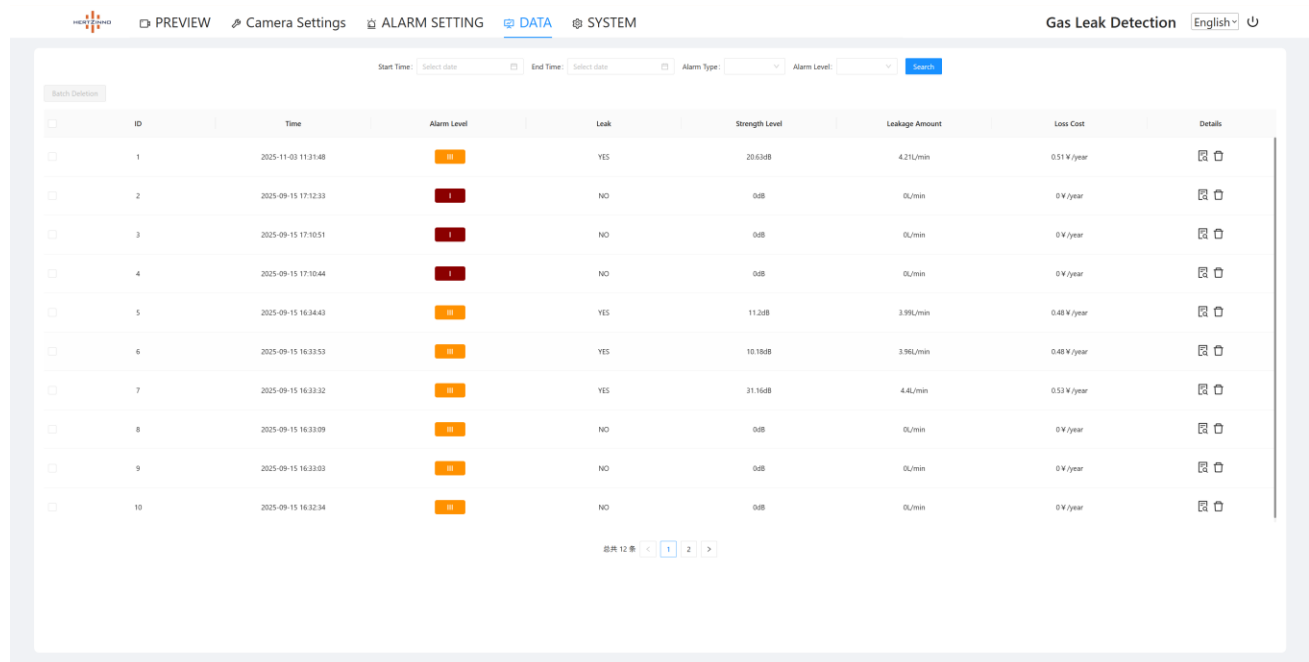
The screenshot shows a web application interface for "Gas Leak Detection". The top navigation bar includes links for PREVIEW, Camera Settings, ALARM SETTING (active), DATA, and SYS. A status message "Deleted Successfully" is visible. The main content area is titled "Alarm Policy Table" and contains a table with the following data:

ID	Policy Name	Related Alarm Conditions	Detection Cycle	Sensitivity	Alarm Level	Action
15	hertzinnno1	hertzinnno1	1s	60%	High	> [Edit] [Delete] [Refresh]

Figure 4-4-1 Alarm Settings

4.5 Data

This page primarily records file information from inspection logs generated during equipment operation. Due to limited storage capacity, only the most recent 100 inspection records are retained. Once this limit is exceeded, the earliest saved records are deleted to make space for new data, enabling cyclic storage. On this page, you can search inspection records by their generation time, view environmental characteristic values at the time of inspection, and access or download images, videos, and other content captured during inspections.



ID	Time	Alarm Level	Leak	Strength Level	Leakage Amount	Loss Cost	Details
1	2025-11-03 11:31:48	NO	YES	20.63dB	4.21L/min	0.51 ¥/year	 
2	2025-09-15 17:52:33	YES	NO	0dB	0L/min	0 ¥/year	 
3	2025-09-15 17:50:51	YES	NO	0dB	0L/min	0 ¥/year	 
4	2025-09-15 17:50:44	YES	NO	0dB	0L/min	0 ¥/year	 
5	2025-09-15 16:34:43	NO	YES	11.3dB	3.99L/min	0.48 ¥/year	 
6	2025-09-15 16:33:53	NO	YES	10.19dB	3.99L/min	0.48 ¥/year	 
7	2025-09-15 16:33:32	NO	YES	31.16dB	4.4L/min	0.53 ¥/year	 
8	2025-09-15 16:33:09	NO	NO	0dB	0L/min	0 ¥/year	 
9	2025-09-15 16:33:03	NO	NO	0dB	0L/min	0 ¥/year	 
10	2025-09-15 16:32:34	NO	NO	0dB	0L/min	0 ¥/year	 

Figure 4-5-1 Data

4.6 Statistic Mapping

To comprehensively understand and analyze the acoustic characteristics of each sound field point throughout a historical day, we need to conduct detailed statistical analysis and visualization of sound pressure information and dominant frequency information at specific times.

- **Sound Pressure Level Statistics:** For each sound field point throughout the historical day, precisely measure and record its sound pressure level at specified times. This data will help us understand the distribution of sound intensity across different time periods at each point, enabling us to assess compliance with relevant noise standards or identify potential acoustic issues.
- **Main Frequency Statistics:** Beyond sound pressure data, we will also compile dominant frequency information for each sound field point at specified times. The dominant frequency represents the frequency component with the highest energy content in a sound signal. It is crucial for identifying the source, nature, and potential acoustic impact of sound signals. By analyzing dominant frequency data, we can further examine the acoustic characteristics of each sound field point, such as the presence of specific noise or vibration sources.



Figure 4-6-1 Statistical Mapping

4.7 Statistical Replay

All the Main frequency and SPL information are logged here.


[PREVIEW](#)
[Camera Settings](#)
[ALARM SETTING](#)
[DATA](#)
[SYSTEM](#)

Gas Leak Detection English 

2025-10-27 11:59:49
2025-11-03 11:59:49

ID	Time	target-15PL	target-25PL	target-35PL	target-45PL	target-55PL	target-65PL	target-1Main Frequency	target-2Main Frequency	target-3Main Frequency	target-4Main Frequency	target-5Main Frequency	target-6Main Frequency
15030	2025-11-03 11:51:31	-	-	-	-	-	-	1160	1160	1160	1160	1160	1160
15034	2025-11-03 11:31:31	-	-	-	12	-	-	0	0	0	0	0	0
15018	2025-11-03 11:11:31	-	-	-	-	-	-	1031	1031	1031	1031	1031	1031
15012	2025-11-03 10:51:31	-	-	-	-	-	-	0	0	0	0	0	0
15006	2025-11-03 10:31:31	-	-	-	-	-	-	773	773	773	773	773	773
15000	2025-11-03 10:11:31	-	-	-	-	-	-	902	902	902	902	902	902
14994	2025-11-03 09:51:31	-	-	-	-	-	-	1031	1031	1031	1031	1031	1031
14988	2025-11-03 09:31:31	26	-	-	-	-	-	0	0	0	0	0	0
14982	2025-11-03 09:11:31	-	-	-	-	-	-	0	0	0	0	0	0
14976	2025-11-03 08:51:31	12	10	-	-	-	-	0	0	0	0	0	0

<
1
2
3
4
5
...
58
>

Figure 4-7-1 Statistical Replay Interface

4.8 System Configuration

Entering the system configuration interface allows you to make changes to device time, network, user management, RTSP management, system information, etc.

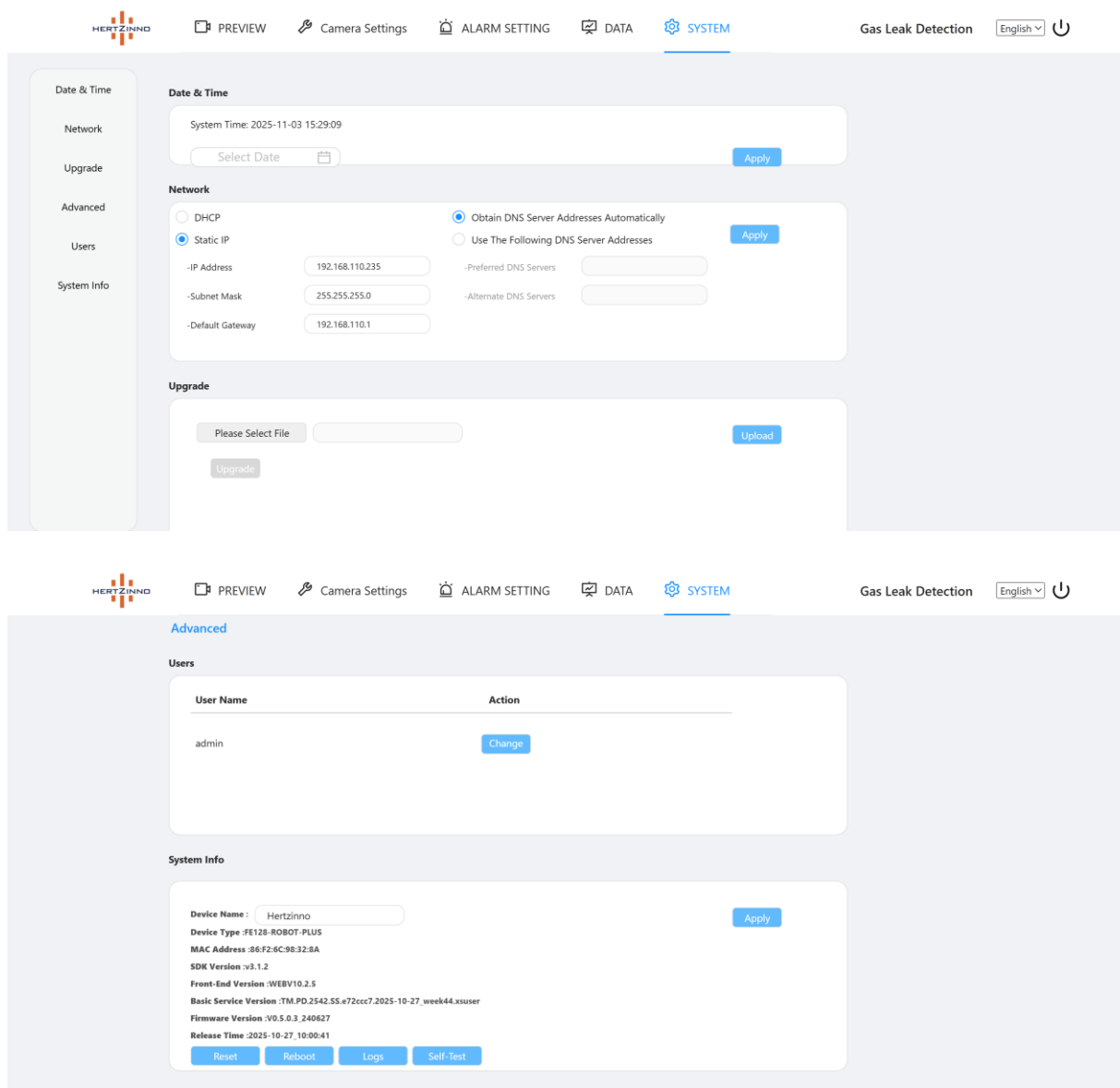
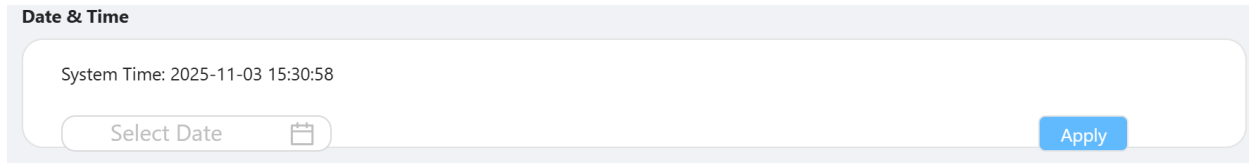


Figure 4-8-1 System Overview

Time Configuration

The system time is the current time of the device. Click the calendar icon, and the user can set the device

time by themselves.

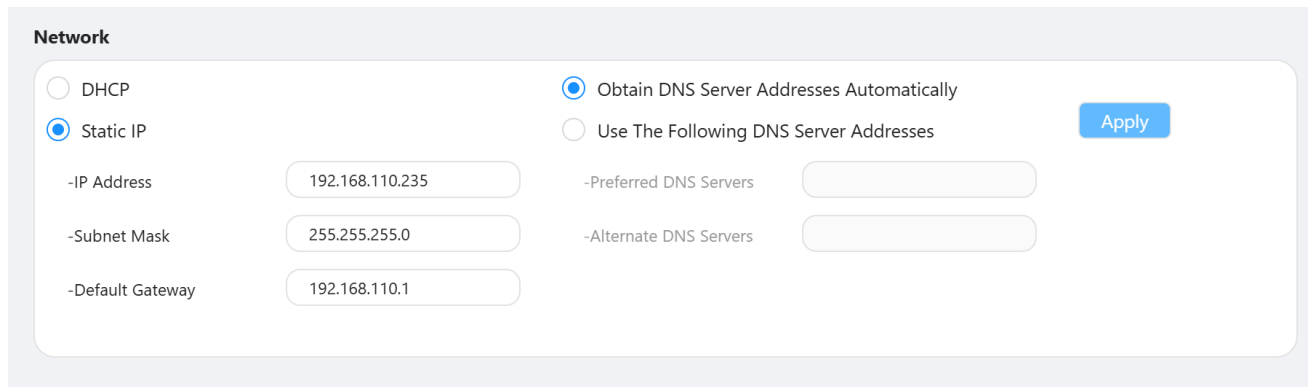


The image shows a 'Date & Time' settings window. At the top, it says 'Date & Time'. Below that, a text box displays 'System Time: 2025-11-03 15:30:58'. At the bottom left, there is a 'Select Date' button with a calendar icon. At the bottom right, there is a blue 'Apply' button.

Figure 4-8-2 Date&Time Settings

Network Configuration

The factory default IP address of the device is 192.168.2.168. If you need to change it, you can easily adjust it through its network configuration interface. Simply enter the corresponding settings option, input the new IP address you wish to set, and click the "Settings" button to complete the modification. Please note that once the settings are completed, the system will automatically disconnect the current login session to ensure the immediate effectiveness of the configuration. At this point, you need to re-enter the newly set IP address in the browser's address bar to log in again and access the system.



The image shows a 'Network' settings window. At the top, it says 'Network'. Below that, there are two radio buttons: 'DHCP' (unselected) and 'Static IP' (selected). To the right of these are two more radio buttons: 'Obtain DNS Server Addresses Automatically' (selected) and 'Use The Following DNS Server Addresses' (unselected). Below the 'Static IP' section, there are three input fields: '-IP Address' (192.168.110.235), '-Subnet Mask' (255.255.255.0), and '-Default Gateway' (192.168.110.1). To the right of these are two more input fields: '-Preferred DNS Servers' and '-Alternate DNS Servers'. At the bottom right, there is a blue 'Apply' button.

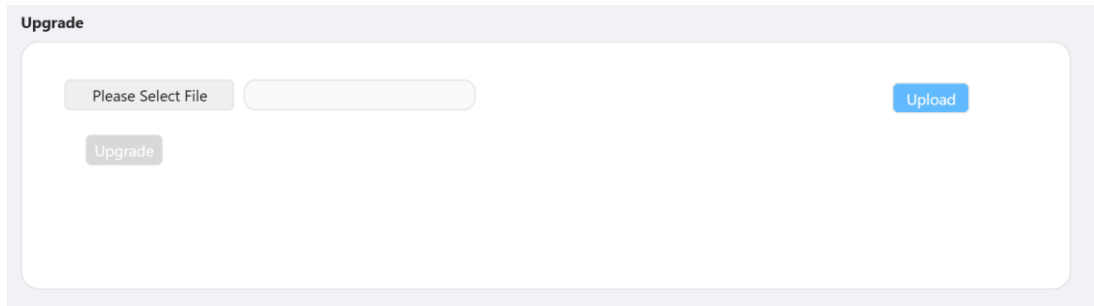
Figure 4-8-3 IP Configuration

System Upgrade

The device supports remote upgrades. When there is a software update available, the obtained update package can be loaded into the device through remote upgrade.

Operation method:

1. Select the update package: First, please click the "Please Select File" button and choose the update package file you wish to use for the upgrade from your local storage.
2. Upload the update package: After selecting the file, please confirm that your selection is correct and click the "Upload" button again. At this point, the system will start uploading the update package file to the device.
3. Perform the upgrade operation: Once the file is uploaded successfully, you will see a "Upload" button become operational. Please click this button, and the device will immediately start the remote software upgrade process.



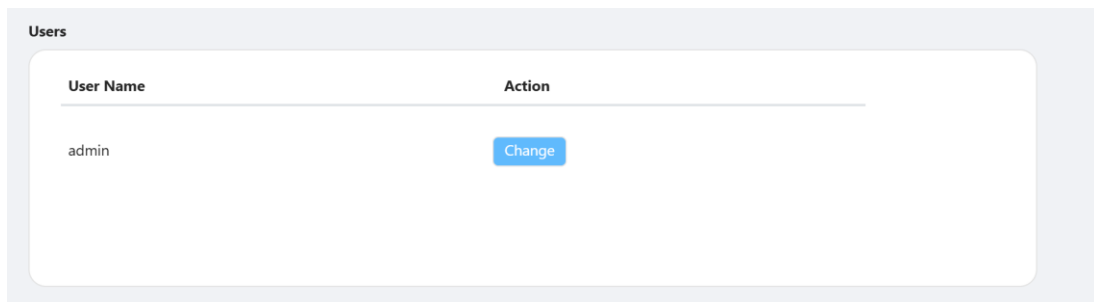
The image shows a web interface titled "Upgrade". It contains a "Please Select File" button, a text input field, an "Upload" button, and an "Upgrade" button.

Figure 4-8-4 System Upgrade

Users

Click "Change", and users can modify their login password by themselves. After modification, just confirm it.

Default account: admin (cannot be modified) Password: 123456.



The image shows a web interface titled "Users". It contains a table with two columns: "User Name" and "Action". The table has one row with the user name "admin" and a "Change" button in the "Action" column.

User Name	Action
admin	Change

Figure 4-8-5 Users Management

Advanced

Advanced settings include RTSP settings and scheduled restart settings.

In the RTSP settings, users are allowed to configure RTSP in detail here. Users have the following permissions:

1. Enable/disable RTSP: Users can freely choose to enable or disable the RTSP service according to actual needs.
2. Streams Number: In different application scenarios, users can customize the maximum number of concurrent connections allowed by RTSP.
3. RTSP Link: To provide a more flexible access method, users have the right to customize the RTSP link address.
4. Currently Connected RTSP Clients: For the convenience of user monitoring and management, the system displays the current number of RTSP connections in real-time.

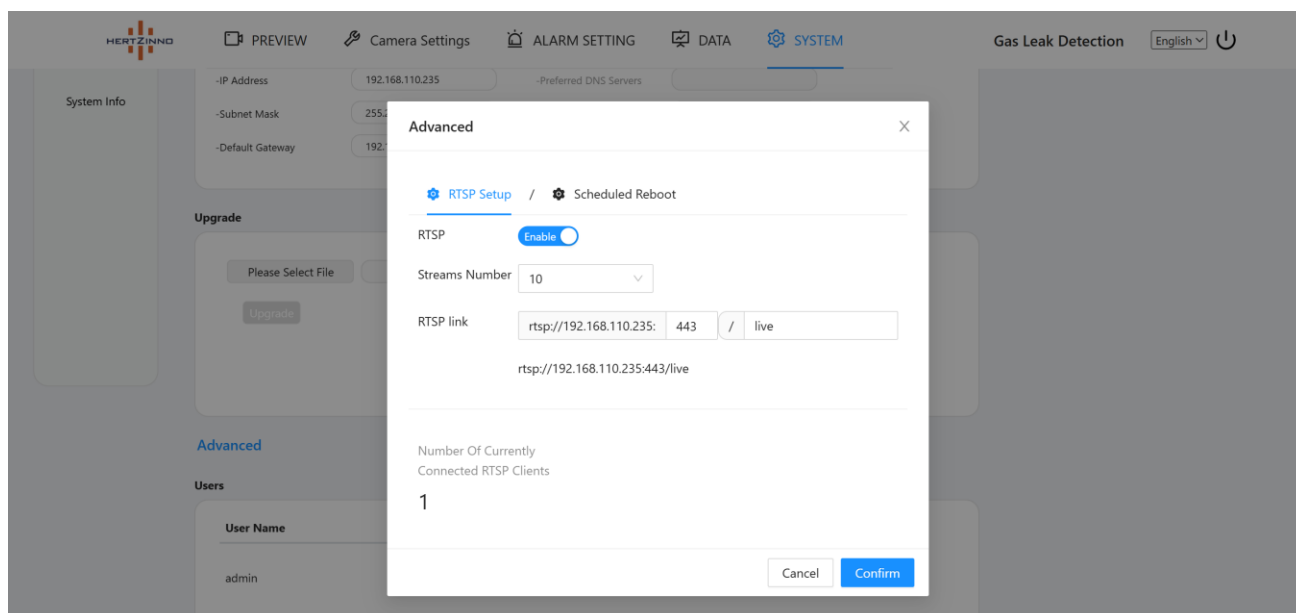


Figure 4-8-6 RTSP Setup Interface

In the scheduled restart settings, users are allowed to restart at a scheduled time or at time intervals.

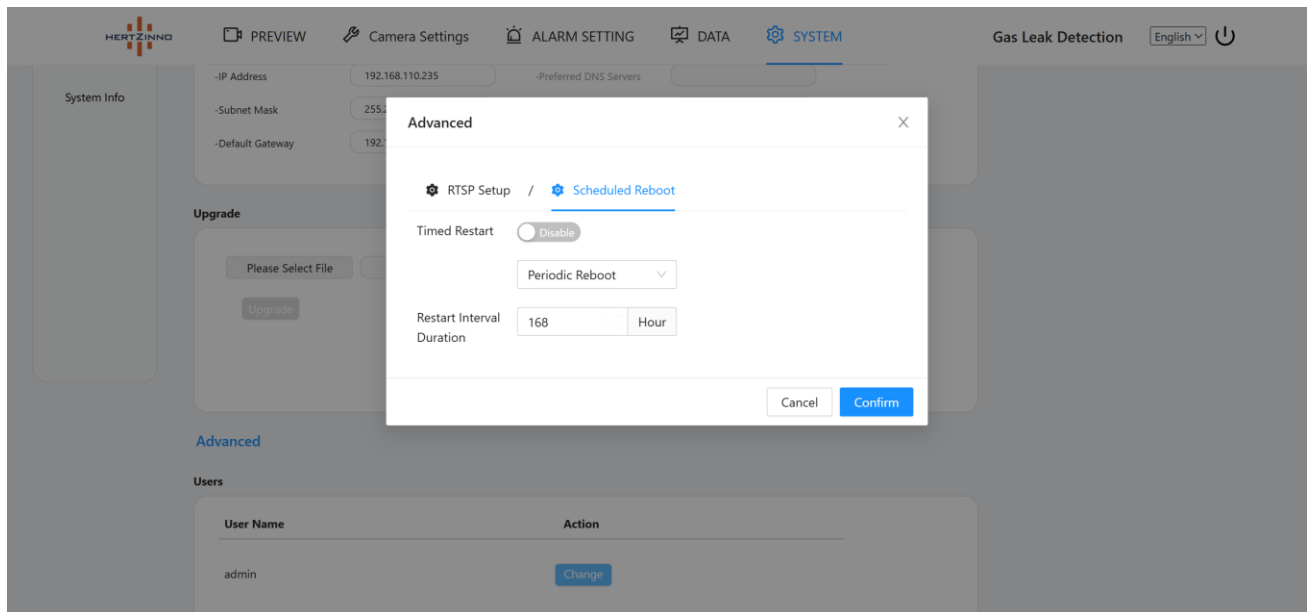


Figure 4-8-7 Scheduled Reboot

System Information

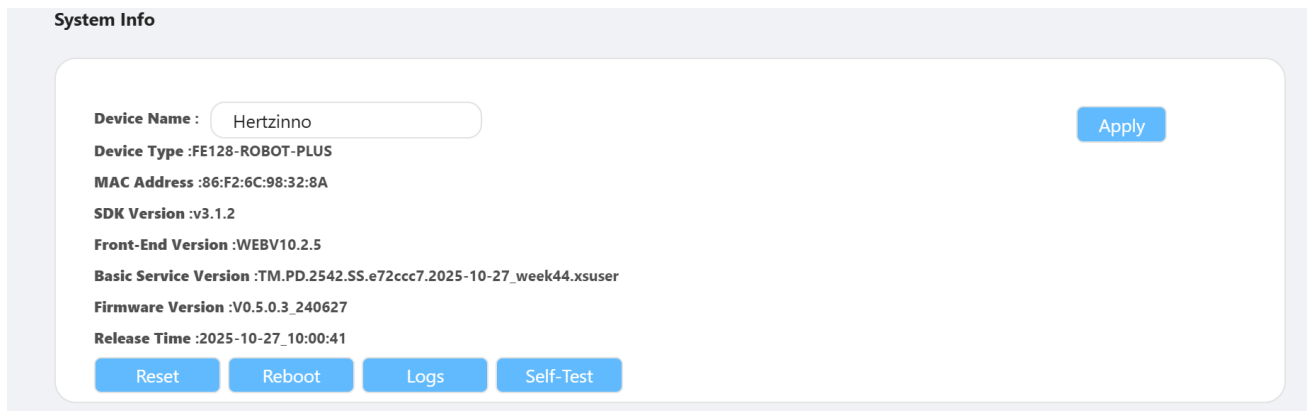
Display current device information and support custom naming of the device.

Reset: One-click restoration of default settings. Information such as IP, username, and password will be reset to default values, and all historical data will be deleted.

Reboot: Click the restart device button to remotely restart the device. After restarting, the system will automatically exit. You need to re-enter the IP address in the browser to enter the system after the device starts automatically.

Logs: Click to set logs, and the system will automatically download the log file locally. The file name is system_logs.log.

Self-test: It can automatically detect whether the hardware microphone is working properly.



The screenshot shows a web interface titled "System Info". It contains the following information and controls:

- Device Name :** A text input field containing "Hertzinno".
- Device Type :** FE128-ROBOT-PLUS
- MAC Address :** 86:F2:6C:98:32:8A
- SDK Version :** v3.1.2
- Front-End Version :** WEBV10.2.5
- Basic Service Version :** TM.PD.2542.SS.e72ccc7.2025-10-27_week44.xsuser
- Firmware Version :** V0.5.0.3_240627
- Release Time :** 2025-10-27_10:00:41
- Buttons:** An "Apply" button is located to the right of the "Device Name" field. At the bottom, there are four buttons: "Reset", "Reboot", "Logs", and "Self-Test".

Figure 4-8-8 System Information

5 API Demonstration

5.1 HTTP Interface

Token Authentication Retrieval

Function Description: Token distribution interface. Verifies username and password to return an encrypted token value.

All subsequent interfaces require the token value to be included in the Authorization header for authentication. If authentication fails or the token is invalid, a 401 error is returned.

Request Format: http://localhost/login

Request Method and URL: POST /login

Request Body: Required

Headers:

Parameter Name	Parameter Value	Required
Content-Type	application/JSON	Required

Encrypted Request Example:

```
{  
  
  "username": "admin",  
  
  "password": "*****"  
}
```

API Request Parameter Description:

Parameter Name	Type	Description
username	string	Username
password	string	Password + LF XS MD5-encrypted ciphertext, 32 characters lowercase

The interface returns JSON data as follows:

```
{
  "token": "*****"
}
```

Retrieve Acoustic Field Configuration

Function Description: Used to retrieve parameters such as the MIC matrix sampling rate (in kHz), upper limit of the analysis frequency band (in kHz), and lower limit of the analysis frequency band (in kHz).

Request Format: http://localhost/system/acoustic

Request Method and URL: GET /system/acoustic

Request Body: None

Return JSON data as follows:

```
{
  "freq_up_limit": 25,
  "freq_down_limit": 20,
  "samplerate": 13200,
  "ret": 200,
```

```
"msg": "SUCCESS",
}
```

Interface Response Parameter Description

Parameter Name	Type	Required	Description
freq_up_limit	integer	Required	Upper limit of the analysis frequency band (in kHz)
freq_down_limit	integer	Required	Lower limit of the analysis frequency band (in kHz)
samplerate	integer	Required	MIC matrix sampling rate (in kHz)
ret	integer	Required	Return Code
msg	String	Required	Return Information

Set Sound Field Configuration

Function Description: Used to set parameters such as MIC matrix sampling rate (in kHz), upper limit of analysis frequency band (in kHz), and lower limit of analysis frequency band (in kHz).

The upper and lower limits of the analysis frequency band adjust the current monitoring range. The adjustable range is 2–65 kHz, and input values must be integers. The difference between upper and lower limits cannot exceed 20 kHz or fall below 2 kHz, and the upper limit must exceed the lower limit. The MIC matrix sampling rate supports 96 kHz and 132 kHz; modification during operation is not recommended.

Request Format: <http://localhost/system/acoustic>

Request Method and URL: PUT /system/acoustic

Request Body: Required

Headers:

Parameter Name	Parameter Value	Required
Content-Type	application/JSON	Required

Request Example:

```
{
  "freq_up_limit": 25,
  "freq_down_limit": 20,
  "samplerate": 132000,
}
```

API Request Parameter Description

Parameter Name	Type	Required	Description
freq_up_limit	integer	Optional	Upper limit of analysis frequency band (in kHz) (4 to 65, integer)
freq_down_limit	integer	Optional	Lower limit of analysis frequency band (in kHz) (integer between 2 and 63)
samplerate	integer	Optional	MIC matrix sampling rate (in kHz) (96000 or 132000)

Retrieve imaging configuration

Function Description: Used to obtain values for parameters such as energy threshold and number of detected targets.

Request Format: http://localhost/system/image

Request Method and URL: GET /system/image

Request Body: None

ReturnedJSON data as follows:

```
{
  "energy_threshold": 25,
  "target_number": 1,
  "ret": 200,
  "msg": "SUCCESS",
}
```

Interface Response Parameter Description

Parameter Name	Type	Required	Description
energy_threshold	integer	Required	Imaging threshold
target_number	integer	Required	Target count
imaging_sensitivity	integer	Required	Imaging filter sensitivity (filter level), valid range [0,2]
imaging_enhance	integer	Required	Imaging enhancement, valid

			range [0,2], where 0 indicates disabled
ret	integer	Required	Return code
msg	String	Required	Return message

Set imaging configuration

Function Description: Used to set values for parameters such as imaging thresholds and target detection counts.

The imaging threshold adjusts the detection threshold value. When the ambient energy level exceeds the threshold, the status changes to abnormal, enabling real-time acoustic field imaging. The video displays an acoustic field imaging marker. The maximum number of detectable targets is currently 3.

Request Format: `http://localhost/system/image`

Request Method and URL: `PUT /system/image`

Request Body: Required

Headers:

Parameter Name	Parameter Value	Required
Content-Type	application/JSON	Required

Request Example:

```
{
  "energy_threshold": 25,
```

```
"target_number": 1,

}
```

Interface Request Parameter Description

Parameter Name	Type	Required	Description
energy_threshold	integer	Required	Imaging threshold
target_number	integer	Required	Target count
imaging_sensitivity	integer	Required	Imaging Filter Sensitivity (Filter Level), valid range [0,2]
imaging_enhance	integer	Required	Imaging enhancement, valid range [0,2], where 0 indicates disabled

Get Video & RTSP Configuration (Video Parameters)

Function Description: Used to retrieve values for parameters such as resolution, video frame rate, video bitrate, and video flip.

Request Format: http://localhost/system/video

Request Method and URL: GET /system/video

Request Body: None

Returned JSON data as follows:

```
{

  "rtsp_client_limit": 6,

  "rtsp_connected_num": 1,
```

```

"rtsp_enabled": 1,

"video_bit_rate": 4096,

"video_flip_model": 0,

"video_fps": 30,

"video_height": 1080,

"video_port": 443,

"video_url": "live",

"video_width": 1920,

"msg": "SUCCESS",

"ret": 200,

}

```

Interface Response Parameter Description

Parameter Name	Type	Required	Description
video_bit_rate	integer	Required	Video bitrate
video_fps	integer	Required	Video Frame Rate
video_height	integer	Required	High resolution
video_width	integer	Required	Resolution Width
video_flip_model	integer	Required	Video Flip Model
video_port	integer	Required	rtsp_server_port
video_url	integer	Required	rtsp service url

rtsp_connected_num	integer	Required	Number of connected RTSP clients
rtsp_client_limit	integer	Required	Maximum number of RTSP clients
rtsp_enabled	integer	Required	RTSP service enabled, 0-disabled, 1-enabled
ret	integer	Required	
msg	String	Must	

Note: Fixed resolution values: 1920*1080, 1280*720

Set Video & RTSP Configuration (Video Parameters)

Function Description: Used to set values for parameters such as resolution, video frame rate, video bitrate, and video flip.

Request Format: http://localhost/system/video

Request Method and URL: PUT /system/video

Request Body: Required

Headers:

Parameter Name	Parameter Value	Required
Content-Type	application/JSON	Required

Example: When setting resolution, the resolution height and width fields must be transmitted simultaneously with fixed values.

```
{
  "rtsp_client_limit": 4,
  "rtsp_connected_num": 1,
  "rtsp_enabled": 1,
  "video_bit_rate": 4096,
  "video_flip_model": 0,
  "video_fps": 30,
  "video_height": 1080,
  "video_port": 443,
  "video_url": "live",
  "video_width": 1920,
}
```

API Request Parameter Description

Parameter Name	Type	Required	Description
video_height	integer	Optional	High resolution
video_width	integer	Optional	Resolution Width
video_fps	integer	Optional	Video frame rate, supports 30
video_bit_rate	integer	Optional	Video bitrate, fixed values: 4096, 2048, 1024, 512
video_flip_model	integer	Optional	Video flip mode: 0: Default display, 1:

			Vertical flip, 2: Horizontal flip, 3: Flip 180°
video_port	integer	Optional	rtsp_server_port
video_url	integer	Optional	rtsp service url
rtsp_client_limit	integer	Non- Required	Maximum number of RTSP clients, valid integer range [1,6]
rtsp_enabled	integer	Non-essential	Enable RTSP service, 0-Disabled, 1-Enabled

Note: Fixed resolution values: 1920×1080, 1280×720

Retrieve operating mode

Function Description: Used to retrieve the current operating mode

Request Format: http://localhost/run_mode

Request Method and URL: GET /run_mode

Request Body: None

ReturnJSON data as follows:

```
{
  "run_mode": 1,
  "ret": 200,
```

```
"msg": "SUCCESS",
```

```
}
```

Interface Response Parameter Description

Parameter Name	Type	Required	Description
run_mode	integer	Required	Detection mode: 1: Single detection mode. -1: Smart patrol mode
ret	integer	Required	Return code
msg	String	Required	Return message

Set operating mode

Function Description: Used to set the operating mode.

Request Format: http://localhost/run_mode

Request Method and URL: PUT /run_mode

Request Body: Required

Headers:

Parameter Name	Parameter Value	Required
Content-Type	application/JSON	Required

Request Example:

```
{
```

```
"run_mode": 1,
```

```
}
```

Interface Request Parameter Description

Parameter Name	Type	Required	Description
run_mode	integer	Required	Detection mode: 1: Single detection mode. -1: Smart cruise mode

Retrieve leak volume estimation configuration (leak mode)

Function Description: Used to obtain parameter values for leak volume estimation.

Request Format: http://localhost/system/leakage

Request Method and URL: GET /system/leakage

Request Body: None

Returned JSON data as follows:

```
{  
  
  "cost_display": 1,  
  
  leakage_distance: 1,  
  
  "leakage_unit": 1,  
  
  "leakage_unit_str": "l/min",  
  
  "monetary_unit": 1,  
  
  "monetary_unit_str": "CNY",  
  
  "energy_cost": 1,
```

```

    "specific_power": 1,

    "value_of_gas": 1,

    "working_hours": 1

    "ret": 200,

    "msg": "SUCCESS",

}

```

Interface Response Parameter Description

Parameter Name	Type	Required	Description
cost_display	integer	Required	Whether to display the leakage cost estimate on the web interface. 0 -- Do not display, 1 -- Display
leakage_distance	integer	Required	Estimated leakage distance, valid range [1~50], unit: meters
leakage_unit	integer	Required	Estimated leakage rate and display unit, 0--l/min, 1--ml/s
leakage_unit_str	string	Required	Estimated and displayed leakage unit, l/min or ml/s
monetary_unit	integer	Required	Currency unit for leak and loss estimation: 0--¥(CNY)[Renminbi], 1 - £(GBP)[Pound Sterling], 2 -- \$(USD)

			[US Dollar], 3 -- € (EUR) [Euro], 4 -- ₪ (RUB)[Ruble]
monetary_unit_str	string	Required	String specifying the currency unit for leak and loss estimates, including the five units listed above: CNY, GBP, USD, EUR, RUB
value_of_gas	float	Required	Gas value, unit: currency unit/m ³
energy_cost	float	Required	Energy cost, unit: currency unit/kWh
specific_power	float	Required	Specific power, unit: kW/(m ³ /min)
working_hours	integer	Required	Average annual operating hours, unit: h/year
ret	integer	Required	Return code
msg	String	Must	Return Information

Configure leak estimation settings (leak mode)

Function Description: Used to set the values of parameters utilized in leak volume estimation.

Parameter names are the same as above. When requesting configuration for a specific parameter, send the corresponding field separately.

Request Format: `http://localhost/system/leakage`

Request Method and URL: `PUT /system/leakage`

Request Body: Required

Headers:

Parameter Name	Parameter Value	Required
Content-Type	application/JSON	Required

Example:

```
{
  "cost_display": 1,
  "leakage_distance": 1,
  "leakage_unit": 1,
  "monetary_unit": 1,
  "energy_cost": 1,
  "specific_power": 1,
  "value_of_gas": 1,
  "working_hours": 1
}
```

API Request Parameter Description

Parameter Name	Type		Description
cost_display	integer	Optional	Whether to display leakage cost estimates on the web interface. 0 -- Do not display, 1 -- Display
leakage_distance	integer	Optional	Estimated leakage distance, valid

			range[1~50], unit: meters
leakage_unit	integer	Optional	Estimated leakage rate and display unit: 0--l/min, 1--ml/s
monetary_unit	integer	Optional	Currency unit for leak and loss estimation: 0 -- ¥ (CNY) [Renminbi], 1 -- £ (GBP) [Pound Sterling], 2 -- \$ (USD) [US Dollar], 3 -- € (EUR) [Euro], 4 -- ₹ (RUB) [Ruble]
value_of_gas	float	Optional	Gas value, unit: currency unit/m ³
energy_cost	float	Optional	Energy cost, unit: currency unit/kWh
specific_power	float	Optional	Specific power, unit: kW/(m ³ /min)
working_hours	integer	Optional	Average annual operating hours, unit: h/year

Retrieve Detection Area

Function Description: Used to obtain the parameter values for the detection area.

Request Format: http://localhost/system/scan_area

Request Method and URL: GET /system/scan_area

Request Body: None

The interface returns JSON data as follows:

```
{  
  "areas": [  
    {  
      "area_height": 168,  
      "area_width": 240,  
      "detect_area_status": 1,  
      "left_top_x": 12,  
      "left_top_y": 12  
    },  
    {  
      "area_height": 168,  
      "area_width": 240,  
      "detect_area_status": 1,  
      "left_top_x": 257,  
      "left_top_y": 185  
    },  
    {  
      "area_height": 168,  
      "area_width": 240,  
      "detect_area_status": 1,  
      "left_top_x": 502,
```

```
"left_top_y":358

},

{

    "area_height":168,

    "area_width":240,

    "detect_area_status":1,

    "left_top_x":747,

    "left_top_y":531

},

{

    "area_height":168,

    "area_width":240,

    "detect_area_status":1,

    "left_top_x":992,

    "left_top_y":704

}

],

"center_focus":2,

"msg":"SUCCESS",

"ret":200

}
```

Interface Response Parameter Description:

Parameter Name	Type	Required	Description
area_height	integer	Required	Rectangle height
area_width	integer	Required	rectanglewidth
detect_area_status	integer	Required	Detect area drawing status: 0 hidden, 1 displayed
left_top_x	integer	Required	Top-left horizontal coordinate
left_top_y	integer	Required	Vertical coordinate of top-left corner
center_focus	integer	Required	Drawing mode: 1: Center focus mode, 2: Selection mode

Set detection area

Used to configure detection area parameters. Parameter names are as above. All fields must be included in the request.

Request Format: http://localhost/system/scan_area

Request Method and URL: PUT /system/scan_area

Request Body: Required

Headers:

Parameter Name	Parameter Value	Required
Content-Type	application/JSON	Required

Example of interface request parameters:

```
{
  "areas":[
    {"area_height":432,
      "area_width":768,
      "detect_area_status":1,
      "left_top_x":576,
      "left_top_y":324}],
  "center_focus":1,
}
```

Parameter Name	Type	Required	Description
area_height	integer	Required	Rectangle height
area_width	integer	Required	rectanglewidth
detect_area_status	integer	Required	Detect area drawing status: 0 hidden, 1 displayed
left_top_x	integer	Required	Top-left horizontal coordinate
left_top_y	integer	Required	Vertical coordinate of top-left corner
center_focus	integer	Required	Drawing mode: 1: Center focus mode, 2: Selection mode

Request to enable detection

Function Description: Used to initiate detection tasks

In Single Detection Mode, detection runs once after initiation and returns the data ID for this session. Results are viewable post-completion on the data review page. In Smart Patrol Mode, detection cycles through enabled alert policies upon initiation, recording results only upon anomaly detection.

Request Format: http://localhost/turn_on_detect

Request Method and URL: POST /turn_on_detect

Request Body: None

ReturnJSON data as follows:

```
{
  "detect_id": 1,
  "ret": 200,
  "msg": "SUCCESS",
}
```

Interface Response Parameter Description

Parameter Name	Type	Required	Description
detect_id	integer	Optional	Detection ID (returned when single-detection mode is enabled)
ret	integer	Required	Return code -1: Indicates no available policies at present
msg	String	Must	Return Information

Request termination detection

Function Description: Used to terminate detection tasks (currently, all detection results are transmitted asynchronously via WebSocket; thus, the termination request returns no content. See Section 5.2.3 for details)

Request Format: `http://localhost/turn_off_detect`

Request Method and URL: `POST /turn_off_detect`

Request Body: None

Get Current Detection Status (Whether Detection is Active)

Function Description: Used to retrieve the detection status

Request Format: http://localhost/system/detect_status

Request Method and URL: `GET /system/detect_status`

Returned JSON data as follows:

```
{  
  
  "detect_status": 0  
  
}
```

Interface Response Parameter Description:

Parameter Name	Type	Required	Description
detect_status	boolean	Required	Detection status 0: Not detecting 1: Detecting

Retrieve Detection Log

Function Description: Used to retrieve the results of a detection task

Request Format: `http://localhost/detect_logs/{detect_id}`

Request Method and URL: GET /detect_logs

Request Body: Request parameters must be included in the HTTP request's Query Param. Relevant parameters are as follows.

Interface Request Parameter Description:

Parameter Name		Default	Description
detect_id	Optional	1	Detection log ID

The interface returns JSON data as follows:

```
{  
  
  "bin_mic_data_name": "mic_data_2019_06_20_20_30_39_940.bin",  
  
  "data_1": 0.0,  
  
  "data_2": -1.0,  
  
  "data_3": 0.0,  
  
  "data_4": 0.0,  
  
  "detect_mode": 2,  
  
  "end_time": 1561033842,
```



```
"files_status": 1,  
  
"id": 7,  
  
"power_spect_photo_name": "power_spectrum_2019_06_20_20_30_39_940.jpg",  
  
"power_spect_video_name": "power_spectrum_2019_06_20_20_30_39_940.mp4",  
  
"saving_path": "2019_06_20_20_30_39_940/",  
  
"soundmap_intensity": 20.4,  
  
"soundmap_photo_name": "sound_2019_06_20_20_30_39_940.jpg",  
  
"soundmap_video_name": "sound_2019_06_20_20_30_39_940.mp4",  
  
"start_time": 1561033839,  
  
"status": 1,  
  
"voiceprint_photo_name": "voiceprint_2019_06_20_20_30_39_940.jpg",  
  
"voiceprint_video_name": "voiceprint_2019_06_20_20_30_39_940.mp4",  
  
"wav_mic_data_name": "mic_data_2019_06_20_20_30_39_940.wav"  
  
},  
  
"msg": "SUCCESS",  
  
"ret": 200  
  
}
```

Interface Response Parameter Description:

Parameter Name	Type	Description
detect_logs	array	Returns a log group. The specific content

		structure is as shown in the example above.
id	int	Log ID
detect_mode	int	Detection Mode
start_time	string	Detection Start Time
end_time	string	End time of detection
status	int	Detection status
soundmap_intensity	int	Sound Field Intensity
file_status	int	File save status: 0-Deleted 1-Saving
saving_path	string	Start time naming folder Record file save path
power_spect_video_name	string	Spectrum diagram video filename identifier
power_spect_photo_name	string	Identifies the spectrum diagram image file name
soundmap_video_name	string	Identifies the sound field imaging video file name
soundmap_photo_name	string	Identifies the sound field imaging image file name
voiceprint_video_name	string	Identifies the voiceprint video filename
voiceprint_photo_name	string	Identifies the voiceprint image filename
bin_mic_data_name	string	Identifies the audio file name for the sound field, in bin format

wav_mic_data_name	string	Identifies the audio file name for the sound field, in wav format
data_1	number	Leakage amount
data_2	number	Leakage Cost
data_3	number	Leakage Volume Unit
data_4	number	Leakage Cost Unit
alarm_strategy_id	int	strategy_id
alarm_level	int	Alarm Level (same as strategy level)
nc_file	string	Sample file, filename
last_alarm_db	number	Alarm threshold in decibels
last_alarm_df	number	Primary Frequency Alarm Value
last_alarm_nc	number	Sample file difference
is_alarm	number	Alarm Status 0: No 1: Yes
nc_file_path	string	Sample file path
nc_file_fft_photo_name	string	Sample file FFT spectrum

Retrieve detection result file

Function Description: Used to obtain the detection result file

Request Format: `http://localhost/detect_file/{log.saving_path}/{logs.file_name}`

Request Method and URL: GET `/detect_file/{log.saving_path}/{logs.file_name}`

Request Body: None

Example Interface Request Parameters (using log results returned in Example 5.1.15):

`http://192.168.2.204/detect_file/2023_04_07_14_31_29_133/sound_2023_04_07_14_31_29_133.jpg`

Returns the corresponding video or image file.

Delete Detection Logs

Function Description: Used to delete detection task results

Request Format: `http://localhost/detect_logs`

Request Method and URL: DELETE/detect_logs

Request Body: Required

Headers:

Parameter Name	Parameter Value	Required
Content-Type	application/JSON	Required

Example of interface request parameters:

```
{  
  "id": [2,3]  
}
```

API Request Parameter Description:

Parameter Name	Type	Description	Required
id	string	Array of IDs for logs to be deleted	Optional

1、 Get system time

Function Description: Used to obtain the current system time

Request Format: http://localhost/system_time

Request Method and URL: GET /system_time

Request Body: None

Returned JSON data as follows:

```
{  
  "sys_time": 17220332  
}
```

Interface Response Parameter Description:

Parameter Name	Type	Required	Description
sys_time	number	Required	System Timestamp

Set System Time

Function Description: Used to set the system time

Request Format: http://localhost/time_set_manually

Request Method and URL: PUT /time_set_manually

Request Body: Required

Headers:

Parameter Name	Parameter Value	Required
----------------	-----------------	----------

Content-Type	application/JSON	Required
--------------	------------------	----------

Example of interface request parameters:

```
{
  "time": 12345678,
}
```

Parameter Name	Type	Required	Description
time	number	Required	System timestamp

Retrieve Alert Policy

Function Description: Used to retrieve alert policies

Request Format: http://localhost/alarm_strategy

Request Method and URL: GET/alarm_strategy

Interface Request Parameter Description:

page	integer	Required	Request page number
------	---------	----------	---------------------

Interface Response Parameter Description:

Name	Type	Required	Remarks
alarm_strategy_list	object[]	Required	item Type: object
id	string	Required	

Name	Type	Required	Remarks
└─name	string	Required	Policy Name
└─alias	string	Required	Alias
└─alarm_condition_id	integer	Required	Associated alarm condition ID
└─alarm_condition_name	string	Required	Associated alarm condition name
└─interval_time	integer	Required	Interval Time
└─times	integer	Required	Count
└─timeout_duration	integer	Required	Timeout Duration
└─sensitivity	number	Required	Sensitivity
└─level	integer	Required	Level 0, 1, 2, 3
└─diagnosis	string	Required	Diagnosis Description
└─switch	integer	Must	Switch 1 On 0 Off
page_total	integer	Required	Total Pages

Add New Alert Policy

Function Description: Used to add an alert policy

Request Format: http://localhost/alarm_strategy

Request Method and URL: POST /alarm_strategy

Notes:

1. Each strategy can only be associated with one alarm condition, which is bound to the strategy via the alarm_id field.

2. Alarm strategy logic conditions include sound pressure level alarms, fundamental frequency alarms, and spectrum comparison. You may flexibly select and configure these conditions based on actual requirements to ensure accurate alarm triggering under specific circumstances.

API Request Parameter Description:

Name	Type	Required	Remarks
name	string	Required	Policy name. The name of the policy, which can be 1 to 20 characters long. For example: "Policy One".
alias	string	Required	Alias
alarm_id	integer	Required	Alarm Condition ID
level	integer	Required	Level 1, 2, 3, 4, used to indicate the urgency or importance level of a policy. By setting different alert levels, you can prioritize alerts of varying types, ensuring critical and urgent alerts receive priority handling when resources are limited.
interval_time	integer	Required	Interval time, the duration between two alerts. In smart cruise mode, the sleep interval after a policy runs before entering the next detection.

Name	Type	Required	Remarks
times	integer	Required	times
timeout_duration	integer	Required	Detection Cycle The duration of the policy cycle, adjustable from 1 to 30 seconds. You may modify the detection cycle length according to your specific requirements.
sensitivity	number	Required	Sensitivity 0-1 The sensitivity of the strategy. Measures the system's responsiveness to abnormal signals during a strategy cycle. Higher sensitivity indicates greater responsiveness to abnormal signals.
Diagnosis	string	Required	Diagnosis Description

Interface Return Parameter Description:

ret	integer	Required	Return code
msg	String	Required	Return message

Delete Alert Policy

Function Description: Used to delete an alert policy.

Request Format: http://localhost/alarm_strategy

Request Method and URL: DELETE /alarm_strategy

Request Body: Required

API Request Parameter Description:

Parameter Name	Type	Description	Remarks
integer []	Array	Array of alert policy IDs to be deleted	Example: [1,2]

Interface Response Parameter Description:

ret	integer	Required	Return code
msg	String	Required	Return message

Modifying Alert Policy

Function Description: Used to modify alert policies

Request Format: http://localhost/alarm_strategy

Request Method and URL: PUT /alarm_strategy

Note:

Active policies cannot be modified.

Interface Request Parameter Description:

Name	Type	Required	Remarks
id	integer	Required	ID of the policy to be modified

Name	Type	Required	Remarks
name	string	Required	Policy name. The name of the policy, which can be 1 to 20 characters long. For example: "PolicyOne".
alias	string	Required	Alias
alarm_id	integer	Required	AlarmConditionID
level	integer	Required	Level 1, 2, 3, 4, used to indicate the urgency or importance level of a policy. By setting different alert levels, you can prioritize alerts of varying types, ensuring critical and urgent alerts receive priority handling when resources are limited.
interval_time	integer	Required	Interval time, the duration between two alerts. In smart cruise mode, the sleep interval after a policy runs before entering the next detection.
times	integer	Required	times
timeout_duration	integer	Required	Detection Cycle The duration of the policy cycle, adjustable from 1 to 30 seconds. You can modify the detection cycle length based on your actual requirements.

Name	Type	Required	Remarks
sensitivity	number	Required	Sensitivity 0-1 The sensitivity of the strategy. Measures the system's responsiveness to abnormal signals during a strategy cycle. Higher sensitivity indicates greater responsiveness to abnormal signals.
Diagnosis	string	Required	Diagnosis Description

Interface Return Parameter Description:

ret	integer	Required	Return code -1: The current policy is already open; modification failed.
msg	String	Required	Return Information

Retrieve Enabled Alert Policy

Function Description: Used to retrieve enabled alert policies

Request Format: http://localhost/alarm_strategy_enable

Request Method and URL: GET/alarm_strategy_enable

Response Parameter Description:

Name	Type	Required	Remarks
alarm_strategy_list	object[]	Required	item Type: object
└─id	string	Required	

Name	Type	Required	Remarks
└─name	string	Required	PolicyName
└─alias	string	Required	Alias
└─alarm_condition_id	integer	Required	Associated alarm condition ID
└─alarm_condition_name	string	Required	Associated alarm condition name
└─interval_time	integer	Required	Interval Time
└─times	integer	Required	Count
└─timeout_duration	integer	Required	Timeout Duration
└─sensitivity	number	Required	Sensitivity
└─level	integer	Required	Level 0, 1, 2, 3
└─diagnosis	string	Required	Diagnosis Description
└─switch	integer	Required	Switch 1 On 0 Off

Set alarm policy switch

Function Description: Used to modify the switch status of the alert policy.

Request Format: http://localhost/alarm_strategy_enable

Request Method and URL: PUT /alarm_strategy_enable

Note:

Enabled strategies cannot be modified.

Interface Request Parameter Description:

Name	Type	Required	Remarks
id	integer	Required	ID of the policy to be modified
switch	integer	Required	Switch 1 On 0 Off

Interface Return Parameter Description:

ret	integer	Required	Return code: 200 for success, -1 for exceeding open policy limit
msg	String	Required	Return message

Get alert conditions

Function Description: Used to retrieve alert conditions

Request Format: http://localhost/alarm_condition

Request Method and URL: GET/alarm_condition

Interface Request Parameter Description:

page	integer	Required	Request page number
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Interface Response Parameter Description:

Name	Type	Required	Remarks
alarm_condition_list	object[]	Required	item Type: object
└─ id	integer	Required	Alarm Condition ID

Name	Type	Required	Remarks
└─name	string	Required	Alarm Condition Alias
└─decibel_thr	integer	Required	Decibel threshold value
└─db_arithmetic_operator	integer	Required	dB arithmetic operators (0-4) >, <, =, >=, <=
└─db_switch	integer	Required	dB alarms switch 0 Off 1 On
└─dominant_freq_thr	integer	Required	Dominant frequency alarm threshold
└─df_arithmetic_operator	integer	Required	Primary frequency arithmetic operators (0-4): >, <, =, >=, <=
└─df_switch	integer	Required	Main Frequency Switch 0 Off 1 On
└─df_logical_operator	integer	Required	Main Frequency Logical Operator 0 1&&
└─noise_comparison_file_id	integer	Required	Noise comparison file ID
└─noise_comparison_thr	number	Must	Noise ratio comparison value 0-1
└─nc_switch	integer	Required	Noise ratio switch 0 Off 1 On
└─nc_logical_operator	integer	Required	Noise comparison logical operator 0 , 1&&
└─nc_arithmetic_operator	integer	Required	Noise ratio comparison arithmetic operators >, <

Name	Type	Required	Remarks
notes	string	Required	
page_total	integer	Required	Total Pages

Add Alert Condition

Function Description: Used to add new alert conditions

Request Format: http://localhost/alarm_condition

Request Method and URL: POST /alarm_condition

Notes:

1. Each alarm condition corresponds to a bound policy. One policy constitutes one detection unit.
2. Logic conditions for alarm policies. These include sound pressure level alarms, main frequency alarms, and spectrum comparison. You can flexibly combine and configure these alarm conditions based on actual needs to ensure accurate triggering under specific circumstances.
4. Sound Pressure Level Threshold Alarm: Determined by comparing the detected sound pressure level with your set threshold value. An alarm triggers when the detected value matches the threshold. Adjust the operator and threshold value as needed.
5. Dominant Frequency Threshold Alarm: Determines the dominant frequency threshold for environmental detection. An alarm triggers when the detected dominant frequency matches your set value. Adjust the operator and dominant frequency value as needed.
6. Spectrum Comparison Alarm: Used to compare against a target file's spectrum chart and determine

similarity. An alarm triggers when the similarity between the detected spectrum chart and the target file's spectrum chart matches your set value. You can adjust the operator and similarity threshold according to your actual needs.

7. An alarm condition is defined as follows: The corresponding switch must be enabled for the associated field to take effect.

db_switch			df_switch			nc_switch		
db_arithmetic_operator	decibel_thr	df_logical_operator	df_arithmetic_operator	dominant_freq_thr	nc_logical_operator	nc_arithmetic_operator	noise_comparison_thr	noise_comparison_file_id
>	50dB	&&	>	1000Hz	&&	>	0.1 [10%]	5

8. For comparison files, refer to the newly added comparison file interface.

Interface Request Parameter Description:

Name	Type	Required	Remarks
name	string	Required	Name of the alert condition. Accepts 1 to 20 characters. Example: "Condition One".
decibel_thr	integer	Required	Alarm value in decibels
db_arithmetic_operator	integer	Required	dB arithmetic operators (0-4) >, <, =, >=, <=
db_switch	integer	Required	dB alarm switch 0 Off 1 On
dominant_freq_thr	integer	Required	Dominant frequency alarm threshold
df_arithmetic_operator	integer	Required	Primary frequency arithmetic operators (0-4) > < = >= <=
df_switch	integer	Required	Main Frequency Switch 0 Off 1 On

Name	Type	Required	Remarks
df_logical_operator	integer	Required	Main Frequency Logical Operator 0 1&&
noise_comparison_file_id	integer	Required	Noise comparison file
noise_comparison_threshold	number	Required	Noise comparison value 0-1
nc_switch	integer	Required	Noise ratio switch 0 Off 1 On
nc_logical_operator	integer	Required	Noise ratio comparison logical operator 0 , 1&&
nc_arithmetic_operator	integer	Required	Noise ratio arithmetic operator: 0 indicates >, 1 indicates <
discharge_switch	string	Required	Discharge Enable 0 Off 1 On
discharge_logical_operator	integer	Required	Logical operator 0 1&&
discharge_type	integer	Required	Discharge type (0: corona, 1: suspension, 2: surface, 4: discharge)
Notes	string	Required	Remarks

Interface Return Parameter Description:

ret	integer	Required	Return code
msg	String	Required	Return message

ModifyAlertConditions

Function Description: Used to modify alert conditions

Request Format: http://localhost/alarm_condition

Request Method and URL: PUT /alarm_condition

Note:

Conditions bound to active policies cannot be modified.

Interface Request Parameter Description:

Name	Type	Is it mandatory?	Note
Name	string	Required	Name of the alert condition. Accepts 1 to 20 characters. Example: "Condition One".
decibel_thr	integer	Required	Alarm value in decibels
db_arithmetic_operator	integer	Required	dB arithmetic operators (0-4) >, <, =, >=, <=
db_switch	integer	Required	dB alarm switch 0 Off 1 On
dominant_freq_thr	integer	Required	Dominant frequency alarm threshold
df_arithmetic_operator	integer	Required	Primary frequency arithmetic operators (0-4): >, <, =, >=, <=
df_switch	integer	Required	Main Frequency Switch 0 Off 1 On

Name	Type	Is it mandatory?	Note
df_logical_operator	integer	Required	Main Frequency Logical Operator 0 1&&
noise_comparison_file_id	integer	Required	Noise comparison file
noise_comparison_threshold	number	Required	Noise comparison value 0-1
nc_switch	integer	Required	Noise ratio switch 0 Off 1 On
nc_logical_operator	integer	Required	Noise ratio comparison logical operator 0 , 1&&
nc_arithmetic_operator	integer	Required	Noise ratio arithmetic operator: 0 indicates >, 1 indicates <
discharge_switch	string	Required	Discharge Enable 0 Off 1 On
discharge_logical_operator	integer	Required	Logical operator 0 1&&
discharge_type	integer	Required	Discharge Type (0: Corona, 1: Flashover, 2: Creepage, 4: Discharge)
Notes	string	Required	Remarks

Interface Return Parameter Description:

ret	integer	Required	Return code
msg	String	Required	Return message

Delete Alarm Condition

Function Description: Used to delete an alarm condition.

Request Format: http://localhost/alarm_condition

Request Method and URL: DELETE /alarm_condition

Request Body: Required

Parameter Description:

Parameter Name	Type	Description	Remarks
integer []	Array	Array of alert condition IDs to be deleted	Example: [1,2]

Interface Return Parameter Description:

ret	integer	Required	Return code -1 Deletion failed. Current condition is bound by policy and cannot be deleted.
msg	String	Required	Return Information

Upload Noise Comparison File

Function Description: Used to upload comparison files.

Request Format: http://localhost/upload_nc

Request Method and URL: POST /upload_nc

Request Body: Required

Note:

- 1、 Only one comparison file can be bound to an alert condition.
- 2、 The comparison file is a system-generated bin file.

Headers

Parameter Name	Parameter Value	Required	Example	Remarks
Content-Type	multipart/form-data	Yes		

API Light Parameters:

Parameter Name	Parameter Type	Required	Example	Remarks
nc_file	file	Yes		
notes	text	is		

{ Content-Disposition: form-

data; name="nc_file"; filename="mic_data_2019_06_22_23_26_47_953.bin"

Content-Type: application/octet-stream

}

Interface return parameters:

Name	Type	Required	Default	Remarks	Additional Information
ret	integer	Required			
msg	string	Required			

Get list of comparison files

Function Description: Used to retrieve the list of alignment files

Request Format: http://localhost/nc_filelist

Request Method and URL: GET/nc_filelist

Interface Request Parameter Description:

page	integer	Required	Request page number
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Interface Response Parameter Description:

Name	Type	Required	Remarks
nc_file_list	object[]	Required	Sample file list, default 10 entries
└─id	integer	Required	File ID
└─file_name	string	Required	File Name
└─file_size	integer	Required	File Size
└─samplerate	integer	Required	sampling rate
└─duration	integer	Required	Duration
└─notes	string	Required	Notes

Name	Type	Required	Remarks
page_total	integer	Required	Total Pages

Delete Comparison File

Function Description: Used to delete alert conditions.

Request Format: http://localhost/nc_file

Request Method and URL: DELETE /nc_file

Request Body: Required

Parameter Description:

Parameter Name	Type	Description	Remarks
integer []	Array	Array of IDs for comparison files to be deleted	Example: [1,2]

Interface Return Parameter Description:

ret	integer	Required	Return code -1 Deletion failed. Current condition is bound by policy and cannot be deleted.
msg	String	Required	Return Information

5.2 WebSocket Interface

To ensure real-time data transmission, the WebSocket protocol is used to retrieve device parameters over port 80. After the connection is established, device parameters are obtained according to a custom communication protocol.

Request format: ws://localhost:port

To ensure secure and reliable WebSocket connections, a heartbeat mechanism is implemented. After connecting to the device's WebSocket server, all active clients must send any data (content unrestricted) to the server at least once per minute to maintain the connection. Failure to do so will cause the server to detect the client as inactive and forcibly terminate the connection.

Retrieve FFT Data

Request address: ws://localhost:port/fftData

The device returns FFT data (the first half of the FFT, length 512) approximately five times per second. The data is returned as a 16-bit int binary stream. Simply parse the returned data in byte order.

Retrieving Voiceprint Images

Request URL: ws://localhost:port/voiceprintData

The device returns one frame of voiceprint image data per second on average. The data is returned as an int binary stream. Parse the returned data in byte order.

Retrieve Detection Results

Request URL: ws://localhost:port/detectFeedback

After each detection, the device returns a packet of detection results in JSON string format. Parse the

returned data in byte order.

The returned JSON data is as follows:

```
{  
  
  "detect_log": {  
  
    "bin_mic_data_name": "mic_data_2019_06_20_20_30_39_940.bin",  
  
    "data_1": 0.0,  
  
    "data_2": -1.0,  
  
    "data_3": 0.0,  
  
    "data_4": 0.0,  
  
    "detect_mode": 2,  
  
    "end_time": 1561033842,  
  
    "files_status": 1,  
  
    "id": 7,  
  
    "power_spect_photo_name": "power_spectrum_2019_06_20_20_30_39_940.jpg",  
  
    "power_spect_video_name": "power_spectrum_2019_06_20_20_30_39_940.mp4",  
  
    "saving_path": "2019_06_20_20_30_39_940/",  
  
    "soundmap_intensity": 20.4,  
  
    "soundmap_photo_name": "sound_2019_06_20_20_30_39_940.jpg",  
  
    "soundmap_video_name": "sound_2019_06_20_20_30_39_940.mp4",  
  
    "start_time": 1561033839,  
  
    "status": 1,  

```

```
"voiceprint_photo_name": "voiceprint_2019_06_20_20_30_39_940.jpg",

"voiceprint_video_name": "voiceprint_2019_06_20_20_30_39_940.mp4",

"wav_mic_data_name": "mic_data_2019_06_20_20_30_39_940.wav"

},

"msg": "SUCCESS",

"ret": 200

}
```

Interface Response Parameter Description:

Parameter Name	Type	Description
detect_logs	array	Returns a log group. The specific content structure is as shown in the example above.
id	int	Log ID
detect_mode	int	Detection Mode
start_time	string	Start Time
end_time	string	End time of detection
status	int	Detection status
soundmap_intensity	int	Sound Field Intensity
file_status	int	File save status: 0-Deleted 1-Saving
saving_path	string	Start time naming folder Record file save path

power_spect_video_name	string	Spectrum diagram video filename identifier
power_spect_photo_name	string	Identifies the spectrum diagram image file name
soundmap_video_name	string	Identifies the sound field imaging video file name
soundmap_photo_name	string	Identifies the sound field imaging image file name
voiceprint_video_name	string	Identifies the voiceprint video filename
voiceprint_photo_name	string	Identifies the voiceprint image filename
bin_mic_data_name	string	Identifies the audio file name for the sound field, in bin format
wav_mic_data_name	string	Identifies the audio file name for the sound field, in wav format
data_1	number	Leakage amount
data_2	number	Leakage Cost
data_3	number	Leakage Volume Unit
data_4	number	Leakage Cost Unit
alarm_strategy_id	int	strategy_id
alarm_level	int	Alarm Level (same as strategy level)
nc_file	string	Sample file, filename

last_alarm_db	number	Alarm threshold in decibels
last_alarm_df	number	Main frequency alarm value
last_alarm_nc	number	Sample file difference
is_alarm	number	Alarm Status 0: No 1: Yes
nc_file_path	string	Sample file path
nc_file_fft_photo_name	string	Sample file FFT spectrum

Retrieve noise point information

Request address ws://localhost:port/energyData

The device periodically returns energy value data for noise points in the form of a 32-bit int binary stream. Simply parse the returned data in byte order.

Transmit multiple sets of target point information based on the current target count. Target point information is arranged sequentially. If the target count is 3, the information is arranged as follows: [[Target 1 Information], [Target 2 Information], [Target 3 Information]]. Within each set, data is arranged as: [Energy, Pixel X-coordinate, Pixel Y-coordinate, Glancing Angle, Azimuth].

Retrieve real-time audio data

Request address: ws://localhost:port/pcmData

The device returns PCM data every second at a sampling rate of 48KHz (48000). The data is returned as a float32 binary stream.

Parse the returned data according to byte order.

6 About Us

HERTZINNO is founded on the fundamentals of acoustic and optical sciences, working closely with leading universities and research institutions in acoustics, optics, and electrical engineering. Since our team's inception in 2018, we have been dedicated to developing technologies with high performance, exceptional sensitivity, and outstanding reliability.

To date, we have accumulated over 100 application sectors worldwide. Our solutions serve diverse industries, including power, energy, oil & gas, chemical, manufacturing, mining, and renewable energy, delivering advanced capabilities for preventive maintenance, online monitoring, and quality control.

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